

Environmental Toxicology

The Unexpected Absence of Nickel Effects on a *Daphnia* Population at 3 Temperatures is Correctly Predicted by a Dynamic Energy Budget Individual-Based Model

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Abstract: Recent studies have shown that temperature affects chronic nickel (Ni) toxicity to *Daphnia magna* at the individual (apical) level. However, the effect of temperature on Ni toxicity to *D. magna* at the population level is unknown. The present study investigated whether the effect of temperature on chronic Ni toxicity to *D. magna* assessed on apical endpoints can be extrapolated to the population level. The results of the population experiment showed no consistent Ni effects on total *D. magna* population abundance at 15, 20, and 25 °C, although the Ni concentrations tested were previously reported to significantly reduce reproduction in *D. magna* individuals. This result supports the idea that ecological risk assessment should not extrapolate as such from apical endpoints to the population level. A dynamic energy budget individual-based model (DEB-IBM) was calibrated using apical Ni toxicity data at 15, 20, and 25 °C. The goal was to investigate whether the calibrated DEB-IBM would be able to predict the unexpected absence of effects at the population level and to further investigate the effect of temperature on Ni toxicity to a *D. magna* population. At the population level, the calibrated DEB-IBM correctly predicted the unexpected absence of an effect of Ni on a *D. magna* population. Detailed analysis of simulation output suggests that the predicted lower Ni sensitivity at the population level occurs because Ni-induced mortality is compensated by reduced starvation (less intraspecific competition). Extrapolated median effective concentration (EC50) values for population density predicted that the effect of temperature on Ni toxicity to *D. magna* populations was smaller (1.9-fold higher at 25 °C than at 15 °C) than on Ni toxicity to *D. magna* apical reproduction (the EC50 is 6.5-fold higher at 25 °C than at 15 °C). These results show that the DEB-IBM can help to replace population experiments by in silico simulations and to optimize the experimental design of population studies. *Environ Toxicol Chem* 2019;38:1423–1433. © 2019 SETAC

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INTRODUCTION

In aquatic ecosystems, where 95% of the species are ectothermic, temperature plays a prominent role (Willmer et al. 2004). The external temperature determines the body temperature of ectothermic organisms, which in turn determines all physiological rates (Paul et al. 2004; Walsh et al. 2014). Population dynamics are also determined by temperature (Pratt 1943; Jiang et al. 2014). For instance, Pratt (1943) showed that maximum *Daphnia magna* population abundance was higher at

18 than at 25 °C and that population structure differed among temperatures.

An increase of temperature is commonly associated with an increase of metal toxicity (Heugens et al. 2001; Sokolova and Lannig 2008). However, the majority of the available studies have focused on acute toxicity. In addition, studies on the effect of temperature on chemical toxicity have, to our knowledge, only assessed apical endpoints (i.e., endpoints that are directly measurable from in vivo toxicity tests, such as mortality, reproductive failure, and reduced growth of single individuals).

An important goal of ecological risk assessment (ERA) is to protect populations against chemical exposure, but ERA is still most commonly based on apical endpoints (Preuss et al. 2009; Galic et al. 2010; Jager et al. 2014). However, previous studies have shown that effect concentrations estimated based on

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apical endpoints may not reflect the effects at the population level (Gagneten and Vila 2001; Viaene et al. 2015).

Recent studies have shown that temperature affects chronic nickel (Ni) toxicity to *D. magna* on apical endpoints (Pereira et al. 2017, 2018). The effect concentration that reduced reproduction by 50% (EC50) increased 2.0- to 6.5-fold with an increase of temperature from 15 to 25 °C, indicating lower toxicity at higher temperatures (Pereira et al. 2017, 2018). Furthermore, in Pereira et al. (2018), the multigenerational Ni effects on *D. magna* reproduction showed different patterns at different temperatures. In addition, in Pereira et al. (2018), the EC10 estimated in the first generation was considered to be protective for *D. magna* in a long-term context. However, the EC10, most commonly used for chronic ERA, was based on apical endpoints, thus disregarding the population-level effects. Therefore, in the present study, we investigated whether the effect of temperature on chronic Ni toxicity to *D. magna* assessed on apical endpoints could be extrapolated to the population level. A population experiment was performed with *D. magna* exposed to 3 Ni concentrations, 0 (control), 12, and 100 µg L⁻¹, previously reported to significantly reduce the apical endpoint reproduction (Pereira et al. 2018).

In a population, organisms compete for food and can be limited by other factors (i.e., intraspecific interaction), such as crowding and light (Preuss et al. 2009; Martin et al. 2013a; Viaene 2016). Daphnids can detect the presence of other individuals through the presence of metabolites or pheromones released into the environment (Gust et al. 2016), which can influence their growth and reproduction (Nishikawa and Ban 1998). Therefore, the fitness of an organism and its response to chemical stressors not only depend on the temperature effect but also may depend on population density (Jiang et al. 2014; Gust et al. 2016).

Differences between the Ni effects on apical reproduction and the Ni effect at the population level were expected, but it was also expected that at 100 µg/L Ni, a concentration that reduced apical reproduction by 70, 50, and 20% at 15, 20, and 25 °C, respectively (Pereira et al. 2018), effects at the population level would be observed. However, the experiment in the present study showed an absence of any consistent Ni effects on total *D. magna* abundance and population structure at 15, 20, and 25 °C (see Results). To explore the possible reasons driving these different results, the effect of temperature on Ni toxicity to a *D. magna* population was studied with a population model: a dynamic energy budget individual-based model (DEB-IBM; Martin et al. 2013a). The DEB-IBM is a generic individual-based implementation of the dynamic energy budget (DEB) theory (Martin et al. 2012). The IBM describes population dynamics through individual behavior, whereas the DEB describes physiological processes on an individual level. In IBM, each organism is generated (at birth) and simulated as an individual entity within the population (Galic et al. 2010). The DEB will translate environmental conditions to metabolic functions (i.e., energy assimilation, maintenance, growth, and reproduction; Kooijman 2009). For each organism, these DEB processes are calculated based on a set of differential equations. The DEB-IBM thus links individual metabolism to

population dynamics. The DEB-IBM for *D. magna* can extrapolate the effect of toxicants measured at apical endpoints to the effects on population dynamics (Martin et al. 2013b).

We calibrated the DEB-IBM using the apical toxicity data for *D. magna* and Ni at different temperatures that were reported in Pereira et al. (2018). We used the DEB-IBM to simulate *D. magna* populations and compared model predictions to observations from the population experiment made in the present study. The Ni concentrations in the population experiment were based on the effect levels reported by Pereira et al. (2018) at the individual level. Considering that these concentrations showed significant effects on reproduction, the initial goal of the experiment was to examine if these effects would propagate to the population level and if the temperature effect on *D. magna* Ni sensitivity would still be present at the population level. In addition, we wanted to test whether the DEB-IBM, calibrated on apical toxicity data at different temperatures, would be able to predict and help explain the absence of effects of Ni at the population level.

MATERIAL AND METHODS

D. magna population experiment

Organism cultures and test media. The *D. magna* clone K6 was cultured in aquaria in a modified M4 medium (Pereira et al. 2017). This medium was also used for exposure. Prior to actual Ni exposure, organisms were acclimated for 2 generations to each temperature treatment (Mitchell and Lampert 2000). More detailed information about the acclimation process and test medium can be found in Pereira et al. (2017).

Experimental design. A *D. magna* population was exposed to 3 Ni concentrations at 15, 20, and 25 °C. The 3 Ni concentrations tested were 0 (control); 12 (nominal concentration), which we expected would produce an approximately 30% effect on *D. magna* reproduction at 15 °C and 0% effect at 20 and 25 °C; and 100 µg L⁻¹ (nominal concentration), which we expected would result in an approximately 50% effect at 20 °C, 20% at 25 °C, and 70% at 15 °C. These conditions and expectations follow the results of previous conventional reproductive toxicity testing (Pereira et al. 2018).

The population experiment started with 6 egg-carrying females that had already released their third or fourth brood and 10 neonates (<24 h) per replicate. Four replicates were performed for each Ni treatment at each temperature treatment. Organisms were exposed in 2 L of medium in aquaria that were continuously aerated. Daphnids were fed daily with *Raphidocelis subcapitata* (formerly known as *Pseudokirchneriella subcapitata*) with a food density of 2.5 mg C L⁻¹ in the first week and 5 mg C L⁻¹ in the following weeks. To consider the different generation time (i.e., time to first brood) observed at the different temperature treatments (Paul et al. 2004; Walsh et al. 2014; Bae et al. 2016), the experiment was conducted over 9 wk (63 d). Half of the medium was renewed 3 times a week (every Monday, Wednesday, and Friday). The total abundance and the population structure (i.e., number of neonates,

juveniles, and adults) were recorded twice a week, every Monday and Thursday. When the number of organisms in the aquaria was above 300, it was necessary to take subsamples to facilitate counting (Wetzel and Likens 2010). Two subsamples of each aquarium were taken after gently spreading the organisms to have a homogeneous distribution of the organisms in the aquarium. The volume of the subsamples was measured and was between 125 and 200 mL. Three population size classes of *D. magna* were quantified by sieving the medium through sieves with different mesh sizes: adults (>800 µm), juveniles (800–500 µm), and neonates (<500 µm; Preuss et al. 2009).

The mean of the 2 subsamples was used to estimate the number of organisms in 2-L aquaria. The coefficient of variation was calculated for all size classes, and if the coefficient of variation was >20%, a third or even a fourth subsample was taken. Every 2 wk (14 d) 50% of each of the 3 size classes were removed at random (i.e., culling) to avoid population collapses attributable to overcrowding (Van Doorslaer et al. 2009). The exposure was performed under a controlled 16:8-h light: dark cycle (Organisation for Economic Co-operation and Development 2012).

Chemical analysis. Dissolved metal concentrations, dissolved organic carbon (DOC), pH, and O₂ were measured once a week, in both fresh and old media. Water samples of fresh and old media of all treatments were collected weekly for analysis of total (only of fresh medium) and dissolved concentrations of Ni, major cations (sodium, potassium, calcium [Ca], and magnesium [Mg]), and DOC. Dissolved concentrations refer to concentrations filtered through a 0.45-µm membrane (Acrodisc Filter, Supor Membrane; PALL). Samples for metal analysis were acidified to a final concentration of 0.14 mol L⁻¹ of nitric acid prior to analysis. Metal analyses were performed using iCAP 7000 Series inductively coupled plasma atomic emission spectroscopy (Thermo Scientific). Samples for DOC analysis were measured with the total organic carbon analyzer (TOC-L CPH; Shimadzu). All data analyses were performed based on the mean dissolved concentrations of new and old media.

The reference material used in all chemical analyses and the quantification limits can be found in Supplemental Data, Table S1. Temperature remained stable in the 3 treatments during the 9 wk of the population experiment (15.2 [±0.6], 19.7 [±1.3], and 25.0 [±1.0] °C; mean ± standard deviation). The DOC (3.5 ± 0.4 mg L⁻¹) and the pH (7.9 ± 0.1) also remained stable during the experiment (Supplemental Data, Table S2). At 20 °C, the measured concentration in the 12-µg Ni L⁻¹ treatment was below the target concentration at 4 sampling time points, but the concentration in the 100-µg Ni L⁻¹ treatment remained stable during the whole exposure (which was 9 wk; Supplemental Data, Table S2). At 15 and 25 °C, the 12- and 100-µg Ni L⁻¹ treatments remained stable during the whole exposure (Supplemental Data, Table S2).

Data analysis—Population experiment. For each endpoint at each counting day and at each temperature, a Dunn's test was applied to test for statistical differences between control

and Ni treatments. The Benjamini and Hochberg (1995) correction method was used to adjust the *p* values for multiple comparisons within each temperature. At 15 °C, reliable data were only available until 21 d of exposure as a result of an experimental error (tap water was mistakenly used as dilution water for media preparation instead of deionized water). The statistical analysis was performed using the dunn.test package (Dinno 2017) in R software (R Development Core Team 2017). Only consistent effects of Ni were considered reliable and used for further data interpretation. A consistent effect of Ni was defined as an effect in the same direction that was found for at least 2 consecutive sampling dates (Van de Perre et al. 2016).

DEB-IBM

An IBM based on the DEB theory for *D. magna* was used to extrapolate effects of Ni to the population level at 15, 20, and 25 °C. An existing DEB-IBM for *D. magna* (Martin et al. 2013a) was calibrated using apical Ni toxicity data (endpoints: reproduction and survival) at 15, 20 and 25 °C (Pereira et al. 2018). The calibrated model was used to predict (blindly) Ni effects at the population level; that is, it was independently validated against the data set from the population experiment. Similar methods have been applied in the past to extrapolate effects of chemicals from the individual to the population level (Martin et al. 2013a; Hochmuth 2016; Viaene 2016).

Model design and calibration of the DEB-IBM. A generic DEB-IBM implementation in NetLogo by Martin et al. (2013a) was developed to predict Ni effects for *D. magna* populations. To simulate the conditions of the population experiment, the original model needed to be adjusted for 1) different temperatures, 2) different food conditions, and 3) the effects of Ni. A brief description of the adjusted DEB-IBM as well as a pseudocode are given in the Supplemental Data.

In the first adjustment the temperature effects were implemented through the Arrhenius equation. The Arrhenius relationship (Equation 1) is used to implement temperature-dependent metabolic rates in the DEB-IBM (Hochmuth 2016; Kooijman 2009). The Arrhenius temperature parameter *T_A* is species-specific and has been determined before for *D. magna* (Kooijman 1988).

$$k(T) = k_1 \times \exp\left(\frac{T_A}{T_1} - \frac{T_A}{T}\right) \quad (1)$$

where *k* is the metabolic rate at temperature *T* (Kelvin), *k₁* is the metabolic rate at the reference temperature *T₁* (293 K), and *T_A* is the Arrhenius temperature (6400 K for *D. magna*; Kooijman 2009).

The second adjustment considers the implementation of correction factors to adjust food-related parameters to account for the differences in algal food and species-specific physiological characteristics (e.g., interclonal differences in *D. magna* in terms of food size preference, food handling time, digestion time, etc.). In the original model, food-related parameters were based on the algae *Desmodesmus subspicatus* as a food

source (Preuss et al. 2009; Martin et al. 2013b). However, in the present study *R. subcapitata* was used as food. To account for such differences, the maximum surface-specific assimilation rate (p_{Am} ; $\text{J d}^{-1}\text{cm}^{-2}$) and the food half-saturation constant (K ; number of cells L^{-1}) were corrected (Kooijman et al. 2008; Equations 2 and 3).

$$K_{\text{food}} = K \times \text{corr}_K \quad (2)$$

$$p_{Am,\text{food}} = p_{Am} \times \text{corr}_{pAm} \quad (3)$$

By multiplying the p_{Am} , we thus implemented a correction factor to take into account differences in energy assimilation efficiency attributable to clone-specific physiological characteristics and food type compared to the original model. In addition, knowing that the food type is different and that feeding characteristics may differ between clones (Repka 1997), the K was corrected as well. Correction factors were determined for each temperature treatment separately because assimilation rates depend on the temperature (Yurista 1999). Optimization of the food correction factors was based solely on the control data of the standard reproduction experiment described by Pereira et al. (2018). The calibrated correction factors for the p_{Am} (corr_{pAm}) were 0.79, 0.84, and 0.76, respectively, for 15, 20, and 25 °C. For K , the correction factors (corr_K) were 0.82, 1.48, and 1.23 for 15, 20, and 25 °C, respectively.

The third adjustment was the estimation of the effect parameters for the different physiological modes of action (PMoAs) to account for Ni toxicity (Martin et al. 2013a). In DEB theory, stressors affect the metabolic rates and thus the energy flows, which is expressed by changing one or more parameters. Depending on the PMoAs, the life history of *D. magna* will be altered. To describe the relationship between the environmental stressor concentration (in this case Ni) and the level of stress on *D. magna*, 2 dose–response models were considered, the hockey-stick and the log-logistic relationships. We finally selected the hockey-stick relationship because this described the effects of Ni on size and reproduction better (Supplemental Data, Figures S1 and S2).

The PMoAs considered were inhibition of assimilation, general maintenance (with an increase in general maintenance costs, i.e., both structural maintenance and maturity maintenance), structural maintenance (increase of structural maintenance costs), maturity maintenance (increase of maturity maintenance costs), growth costs (an increase in costs for growth), and reproduction costs (Martin et al. 2013b; Hochmuth 2016). The models with the different PMoAs (as indicated by Martin et al. 2013a, 2013b) were calibrated with the apical Ni toxicity data of *D. magna* from Pereira et al. (2018). The best-fitting model, that is, the one that showed the lowest euclidean distance (ρ), was further used in the calibrated DEB-IBM.

For calibrating the food correction factors and the effect parameters for Ni toxicity, an approximate Bayesian computation was used (van der Vaart et al. 2016). The model was run for 10 000 iterations, with randomized values of the parameters to be optimized in each iteration. The optimized parameter values were calculated from the 100 best fits based on the ρ between

the model output (predictions) and the apical Ni toxicity data of *D. magna* (endpoints: size and reproduction) from Pereira et al. (2018; observations; Equation 3; van der Vaart et al. 2016).

The ρ is determined as

$$\rho(m_i, D) = \sqrt{\sum_j \left(\frac{m_{i,j} - D_j}{sd(m_j)} \right)^2} \quad (4)$$

where $m_{i,j}$ is the model output of endpoint j in iteration run i , D_j is the observed data of endpoint j , and $sd(m_j)$ is the standard deviation of endpoint j of all simulation runs. Optimal parameter values for each PMoA can be found in Supplemental Data, Table S4.

Lethal effects of Ni are modeled by introducing a stochastic chance of dying that increases with the Ni concentration. Based on Equation 5 from De Laender et al. (2008), the daily mortality rate of a given survival experiment can be calculated as

$$m(\text{Ni}) = \frac{1}{t_{\text{exp}}} \times \ln \left(1 + \left(\frac{[\text{Ni}]}{\text{LC50}} \right)^\beta \right) \quad (5)$$

where LC50 is the 50% lethal concentration, β is the slope of the dose–response curve, and t_{exp} is the duration of the experiment (in days). From the daily mortality rate, the stochastic chance of dying, $p_m(\text{Ni})$, can be calculated as follows:

$$p_m(\text{Ni}) = (1 - e^{-1 \times m(\text{Ni})}) \quad (6)$$

Based on the survival data from Pereira et al. (2018), values for LC50 and β were determined for each temperature (Supplemental Data, Table S5).

Validation of the calibrated DEB-IBM. With the calibrated model (i.e., with the temperature adjustment, optimized food correction factors, and calibrated effect parameters for Ni toxicity), we mimicked the population experiment to know how well the model predicted the observation on population dynamics. The same conditions as the experiment were applied in the model simulation (i.e., daily food addition, medium replacement 3 times per week, and culling every 14 d). The measured dissolved Ni concentrations were imposed on the simulations. The PMoA that best predicted the data of the apical experiment (based on the euclidean distance) was used for these simulations. The predictions of the calibrated DEB-IBM were validated against the (independent) data set from the population experiment at 15, 20, and 25 °C.

Extrapolation of the ECx for a *D. magna* population. The calibrated model was used to further extrapolate the effects of temperature on Ni toxicity to the *D. magna* population and to predict at which Ni concentrations effects would be observed. We again imposed the same conditions as in the population experiment but now varying the Ni concentrations from 0 to 300 $\mu\text{g L}^{-1}$. Population-level effect concentrations were determined for equilibrium population abundance (i.e., the mean population density after 8 and 9 wk of simulations—when the population density reaches a steady state). With the drc

package in R, population-level ECx values were calculated based on regression of a log-logistic dose–response curve (*llogistic*—3-parameter log-logistic curve; Ritz et al. 2015; R Development Core Team 2017).

RESULTS AND DISCUSSION

Population experiment

The population experiment showed no consistent Ni effects on total *D. magna* population abundance at 15, 20, and 25 °C (Figure 1), although the Ni concentrations tested (0, 12, 100 $\mu\text{g L}^{-1}$) were previously reported to significantly reduce the apical endpoint reproduction in a standard reproduction test (Pereira et al. 2018). The tested concentration 12 $\mu\text{g Ni L}^{-1}$ corresponds to an approximately 30% effect on *D. magna* apical reproduction at 15 °C and to 0% effect at 20 and 25 °C (Pereira et al. 2018). The concentration 100 $\mu\text{g Ni L}^{-1}$ corresponds to 70, 50, and 20% effects at 15, 20, and 25 °C, respectively (Pereira et al. 2018). Furthermore, population structure was not affected by Ni: no consistent effect of Ni on the number of neonates, juveniles, and adults was observed at any temperature. Also, no consistent effect of Ni on the ratio between the number of juveniles and the number of adults was observed at any temperature (Figure 2).

Previous studies have also shown that effect concentrations estimated based on apical endpoints may not reflect the effects at the population level (Gagneten and Vila 2001; Viaene et al. 2015). In Viaene et al. (2015), after 1 wk of exposure to 62 $\mu\text{g L}^{-1}$ of pyrene, a concentration reported to have an effect on the apical endpoint survival (EC50 for immobilization = 68 $\mu\text{g L}^{-1}$), the total *D. magna* population abundance was reduced; after 2 wk, the *D. magna* population abundance had begun to recover, but was affected for a longer period of time. The results of Viaene et al. (2015) suggest that apical effect concentrations do not (always) translate to population-level effects in a straightforward way. Another study showed that a *Ceriodaphnia dubia* population became extinct when exposed to 20 $\mu\text{g Cu L}^{-1}$, a concentration that reduced the apical endpoint reproduction by 50% (Gagneten and Vila 2001). To our

knowledge, until now the effect of Ni on a *D. magna* population was only studied by Münzinger (1994) and only at 20 °C in an 8-wk study. In that study, the total abundance of the population held in flow-through cultures was significantly reduced by 120, 160, and 200 $\mu\text{g Ni L}^{-1}$ compared with the control treatment. The study did not, however, report effect concentrations on apical endpoints. Therefore, it is difficult to compare with the present study. The present study and the studies of Viaene et al. (2015) and Gagneten and Vila (2001) demonstrate that very different outcomes (quantitatively) can be obtained for population-level endpoints than for apical endpoints. Thus, conventional ecotoxicological results (i.e., based on apical endpoints) cannot be straightforwardly extrapolated to population-level effects.

During the 9 wk of the experiment the major cation concentrations remained stable at 15 and 25 °C but not at 20 °C (Supplemental Data, Table S3). One could argue that the different major ion concentrations at 20 °C could compromise the comparison of the effect of Ni on the *D. magna* population with that at other temperatures. However, this is unlikely because the biotic ligand model for chronic Ni toxicity to *D. magna* (Deleebeeck et al. 2008) that we applied to media composition only estimated a 1.2-fold difference between the estimated EC10 values of the media at 15, 20, and 25 °C.

DEB-IBM predictions

In the population experiment of the present study the effect of Ni on a *D. magna* population was absent at 15, 20, and 25 °C. Therefore, to further investigate the effect of temperature on Ni toxicity to a *D. magna* population, a DEB-IBM (Martin et al. 2013a) calibrated using apical Ni toxicity data at 15, 20, and 25 °C was used (Pereira et al. 2018).

Physiological modes of action. To predict the population response to Ni, we firstly had to identify the PMoAs which are affected by Ni. The PMoA with an effect on growth costs (an increase in costs for growth) was able to predict the Ni effect on

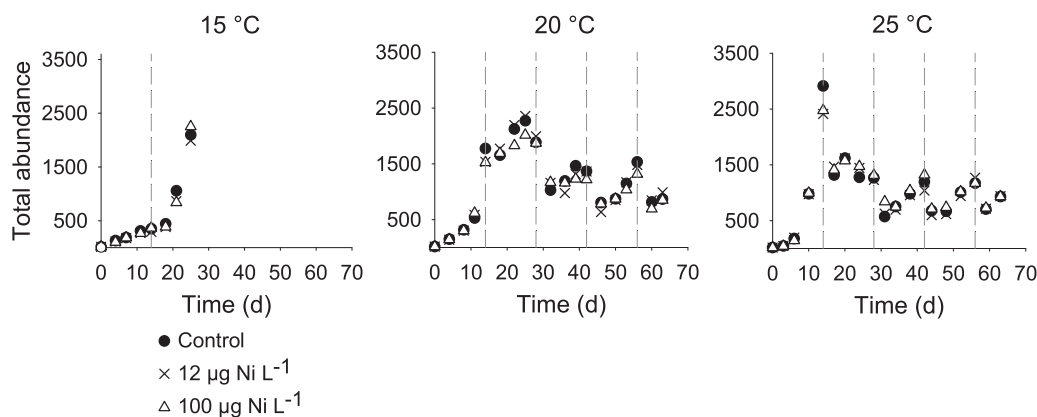


FIGURE 1: Average total abundance of *Daphnia magna* population exposed for 9 wk (63 d) to nickel (Ni; 0, 12, and 100 $\mu\text{g Ni L}^{-1}$) at 15, 20, and 25 °C. Every 14 d half of the population was reduced by random culling. Vertical lines represent the days (every 14 d) that 50% of each of the 3 size classes (i.e., neonates, juveniles, and adults) were removed at random to avoid population collapses attributable to overcrowding. At those time points the total abundance of the *D. magna* population reported refers to the abundance just prior to the culling.

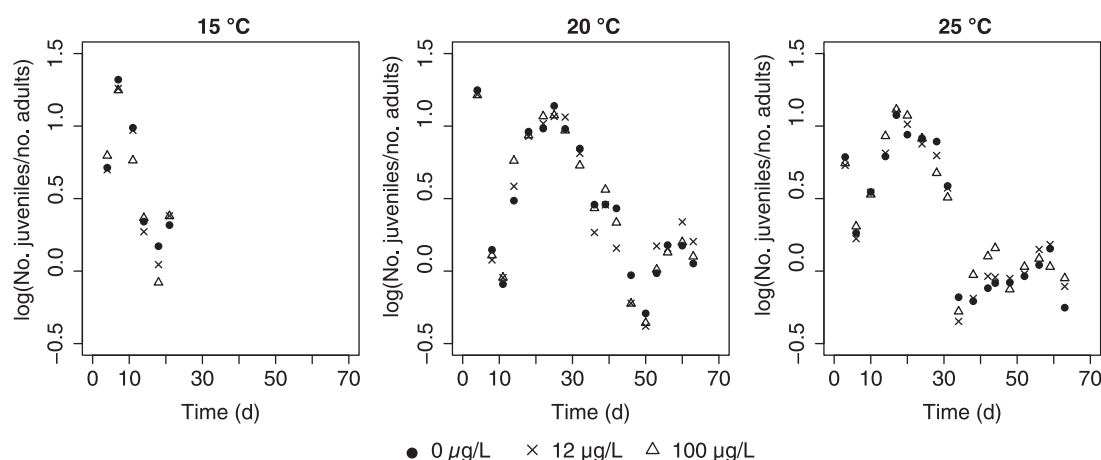


FIGURE 2: Ratio of the number of juveniles and the number of adults of the *Daphnia magna* population exposed for 9 wk (63 d) to nickel (Ni; 0, 12, and 100 $\mu\text{g L}^{-1}$) at 15, 20, and 25 °C.

the apical endpoints size (length) and reproduction observed in Pereira et al. (2018). Furthermore, it presented the lowest ρ and the best fits to the observations (Figures 3 and 4; Supplemental Data, Table S4). The PMoAs with an effect on general maintenance costs (i.e., both structural maintenance and maturity maintenance), on structural maintenance costs, and on assimilation (decrease in feeding ability) also predicted the Ni effect on the apical endpoints size and reproduction relatively well; but the ρ values were considerably higher (Figures 3 and 4; Supplemental Data, Table S4). The PMoAs with an effect on reproduction costs and maturity maintenance costs did not predict an effect on growth. Overall, these results indicate that

the PMoA with an effect on growth costs best predicted the Ni effects on size and reproduction.

The relation between the best-fit PMoA in the present study (increased costs of growth) and the known mechanisms of Ni toxicity is not straightforward. Firstly, the mechanisms of Ni toxicity are not fully understood. Brix et al. (2017) identified as potential mechanisms of Ni toxicity in aquatic organisms the disruption of Ca, Mg, and iron homeostasis. Also, previous studies showed that Ni exposure decreased respiration rates and glycogen and adenosine triphosphate (ATP) concentrations in *Daphnia* (Pane et al. 2003, 2004). Pane et al. (2004) showed that exposure of *D. magna* to Ni decreased Mg

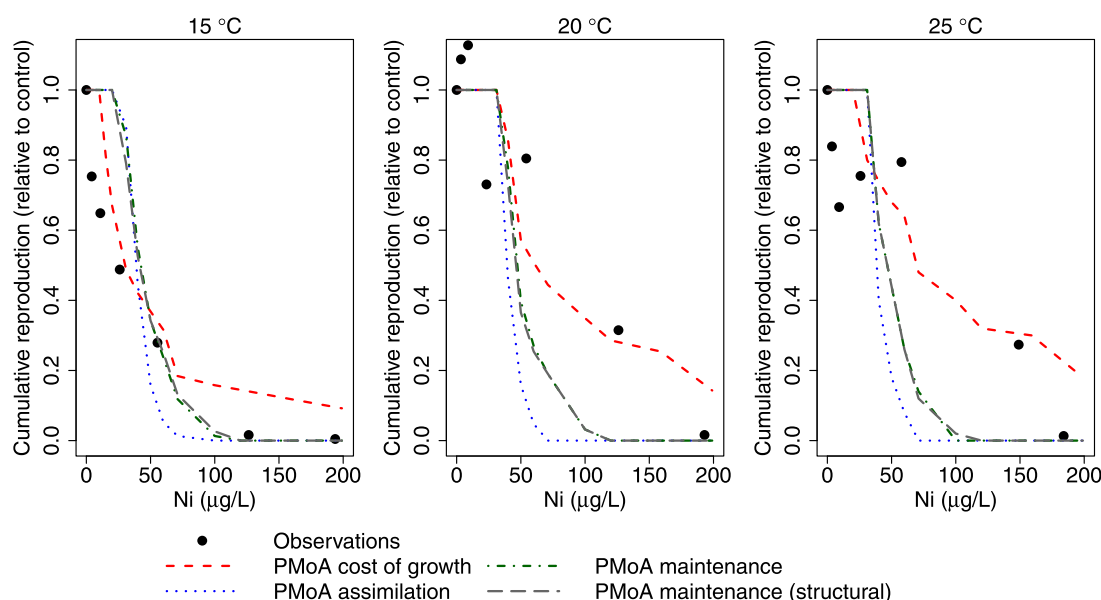


FIGURE 3: Calibration of the dynamic energy budget individual-based model for *Daphnia magna* with different physiological modes of action (PMoAs), that is, an effect on growth costs (an increase in costs for growth), on general maintenance costs (both structural maintenance and maturity maintenance), on structural maintenance costs, and on assimilation (decrease in feeding ability). Predictions are plotted versus the observations from the apical nickel (Ni) toxicity data set from Pereira et al. (2018). The hockey-stick relationship between the Ni concentration and the level of stress on the PMoAs was used. Each graph shows *D. magna* cumulative reproduction as a function of the Ni concentration at 15, 20, and 25 °C. Cumulative reproduction is expressed as the number of offspring per individual female produced until the organisms in the control treatment released the fifth brood as a percentage of the control treatment (cumulative reproduction, relative to control). Lines represent predictions and dots, observations.

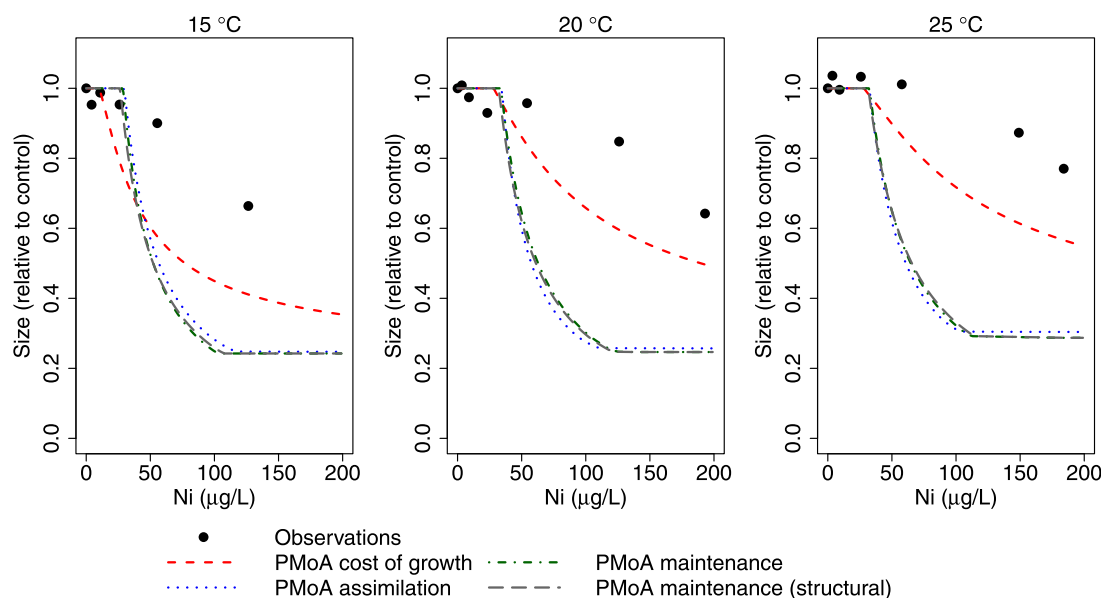


FIGURE 4: Calibration of the dynamic energy budget individual-based model for *Daphnia magna* with different physiological modes of action (PMoAs), that is, an effect on growth costs (an increase in costs for growth), on general maintenance costs (both structural maintenance and maturity maintenance), on structural maintenance costs, and on assimilation (decrease in feeding ability). Predictions are plotted versus the observations from the apical nickel (Ni) toxicity data set from Pereira et al. (2018). The hockey-stick relationship between the Ni concentration and the level of stress on the PMoAs was used. Each graph shows *D. magna* size (as a percentage of control treatment) as a function of the Ni concentration at 15, 20, and 25 °C. Lines represent predictions and dots, observations.

concentrations and reduced ATP levels. That study suggested that those 2 endpoints could be related because Mg^{2+} is a cofactor for ATP (Stryer 1995). The main source of energy in cells, ATP must be bound to an Mg^{2+} ion to be biologically active (Stryer 1995). It would be worthwhile to investigate if the effect of Ni on Mg and ATP levels that are necessary for the growth of organisms (to build new units of structure, e.g., a new cell) could manifest into an increased cost for growth.

The mechanisms of Ni toxicity reported in the previous paragraph could equally well manifest into other PMoAs because ATP is necessary for all biological functions and therefore can have an effect on all PMoAs. Magnesium is essential for several biological functions including structural stabilization of nucleic acid (Wolf and Cittadini 2003), which could relate to the PMoAs of general maintenance costs and structural maintenance costs. Also, Mg is essential for cell membranes and to promote specific structural organization of enzymes (Wolf and Cittadini 2003), which may translate into an effect on PMoA assimilation. Ashauer and Jager (2018) already pointed to the importance of linking the PMoA as defined in the DEB to physiological endpoints (what happens at the cellular level) to enable the extrapolation from in vitro and in vivo toxicity studies to in silico studies.

Predictions versus observations. Overall, the DEB-IBM calibrated to our control condition and, based on the apical Ni toxicity data (Pereira et al. 2018), predicted the absence of Ni effect to a *D. magna* population at 12 and 100 $\mu\text{g Ni L}^{-1}$ at 15, 20, and 25 °C in accordance with the observations from the population experiment (Figures 5–7). Only at 15 °C and 100 $\mu\text{g Ni L}^{-1}$ did there seem to be an underprediction in juvenile, adult, and

total population density. This indicates that there might be a significant effect of Ni at 15 °C at a concentration of 100 $\mu\text{g Ni L}^{-1}$. Unfortunately, because of the lack of reliable data beyond 21 d of exposure, it is not possible to conclude if this trend would have continued. The total abundance of the population over time and the size structure predictions were generally in accordance with the observations (Figures 5–7). Only at 20 °C was the predicted population density at the end of the simulation higher than observed, for all Ni treatments. This is attributable to an overprediction of the juvenile density at 20 °C (Figure 7).

Extrapolation of the Ni EC values for a *D. magna* population at 15, 20, and 25 °C. The extrapolated EC50 for a *D. magna* population of Ni (i.e., using the average of the total population abundance from week 8 to week 9) was lower at 15 °C, 2.4-fold higher at 20 than at 15 °C, and 1.3-fold higher at 20 than at 25 °C (Figure 8 and Table 1). The extrapolated EC50 values indicated that the effect of temperature on Ni toxicity to a *D. magna* population was smaller than the effect of temperature on Ni toxicity to *D. magna* (apical) reproduction. Our previous experiment showed that chronic Ni toxicity to *D. magna* increased with the increase of temperature, with a 6.5-fold difference between the EC50 values for *D. magna* apical reproduction (i.e., 17.3 ± 2.2 , 75.1 ± 8.9 , $129.1 \pm 9.4 \mu\text{g L}^{-1}$ at 15, 20, and 25 °C, respectively; Pereira et al. 2018). The effect of temperature on Ni toxicity to *D. magna* population equilibrium density was thus smaller than that to *D. magna* apical reproduction.

The predicted EC50 values for population equilibrium density are higher than those reported for the apical endpoint of individual reproduction (i.e., a 5.5-fold, 3.1-fold, and 1.4-fold

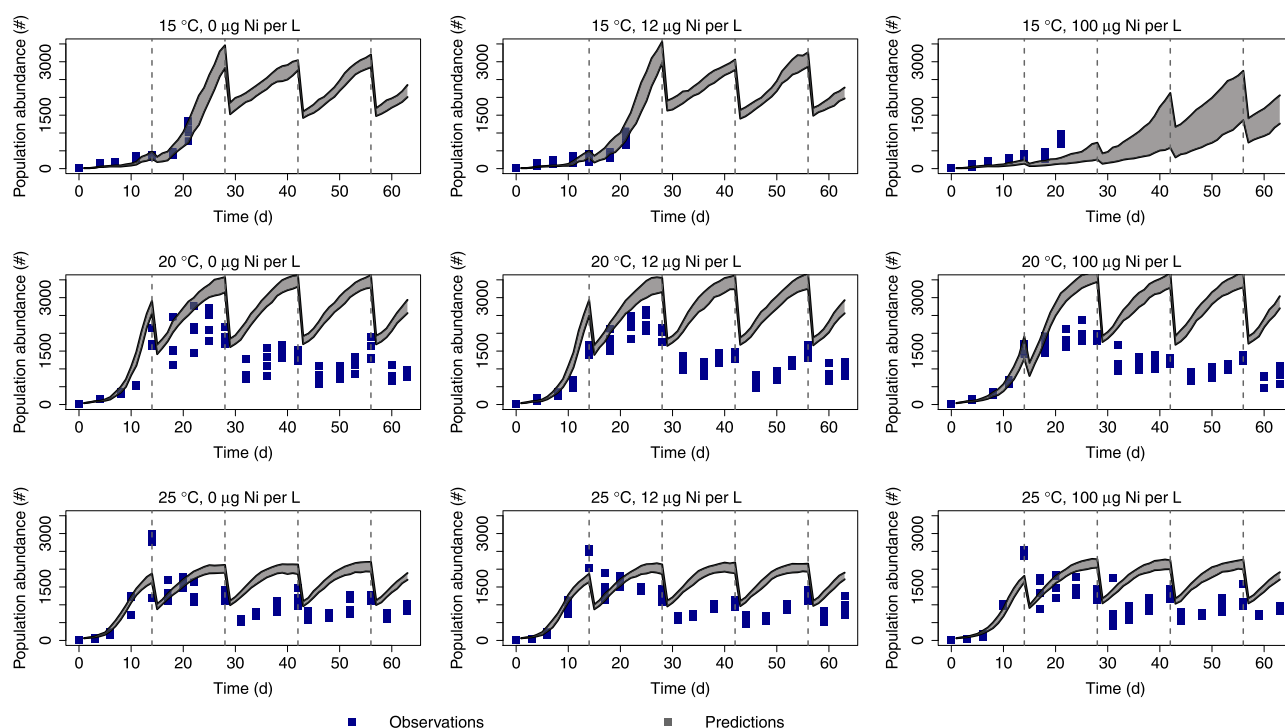


FIGURE 5: Observed and predicted population dynamics for *Daphnia magna* under 0, 12, and 100 μg nickel (Ni) L^{-1} for 9 wk at 15 (data available only until 21 d), 20, and 25 $^{\circ}\text{C}$. Dots represent data points of the total population abundance and lines, the predictions of the model using the physiological mode with an effect on growth costs (an increase in costs for growth). Vertical lines represent the days (every 14 d) that 50% of each of the 3 size classes (i.e., neonates, juveniles, and adults) were removed at random to avoid population collapses attributable to overcrowding.

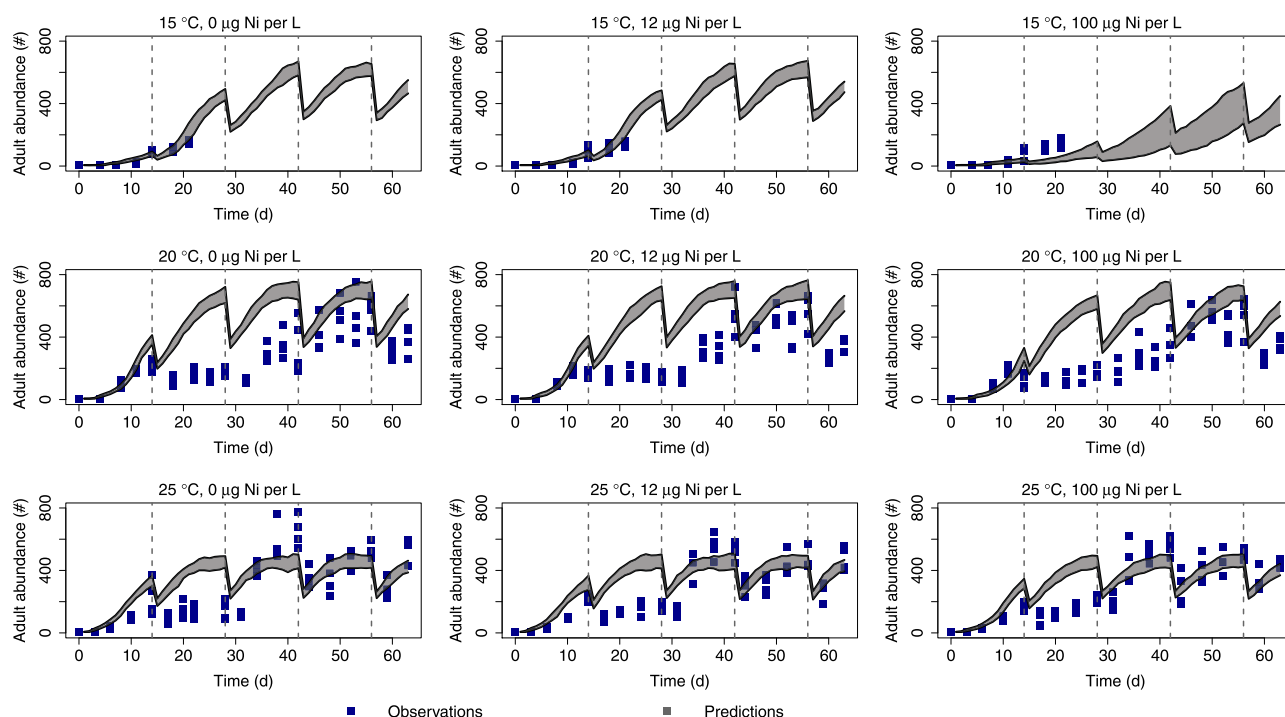


FIGURE 6: Observed and predicted population dynamics for *Daphnia magna* under 0, 12, and 100 μg nickel (Ni) L^{-1} for 9 wk at 15 (data available only until 21 d), 20, and 25 $^{\circ}\text{C}$. Dots represent data points of the adult abundance and lines, the predictions of the model using the physiological mode of action with an effect on growth costs (an increase in costs for growth). Vertical lines represent the days (every 14 d) that 50% of each of the 3 size classes (i.e., neonates, juveniles, and adults) were removed at random to avoid population collapses attributable to overcrowding.

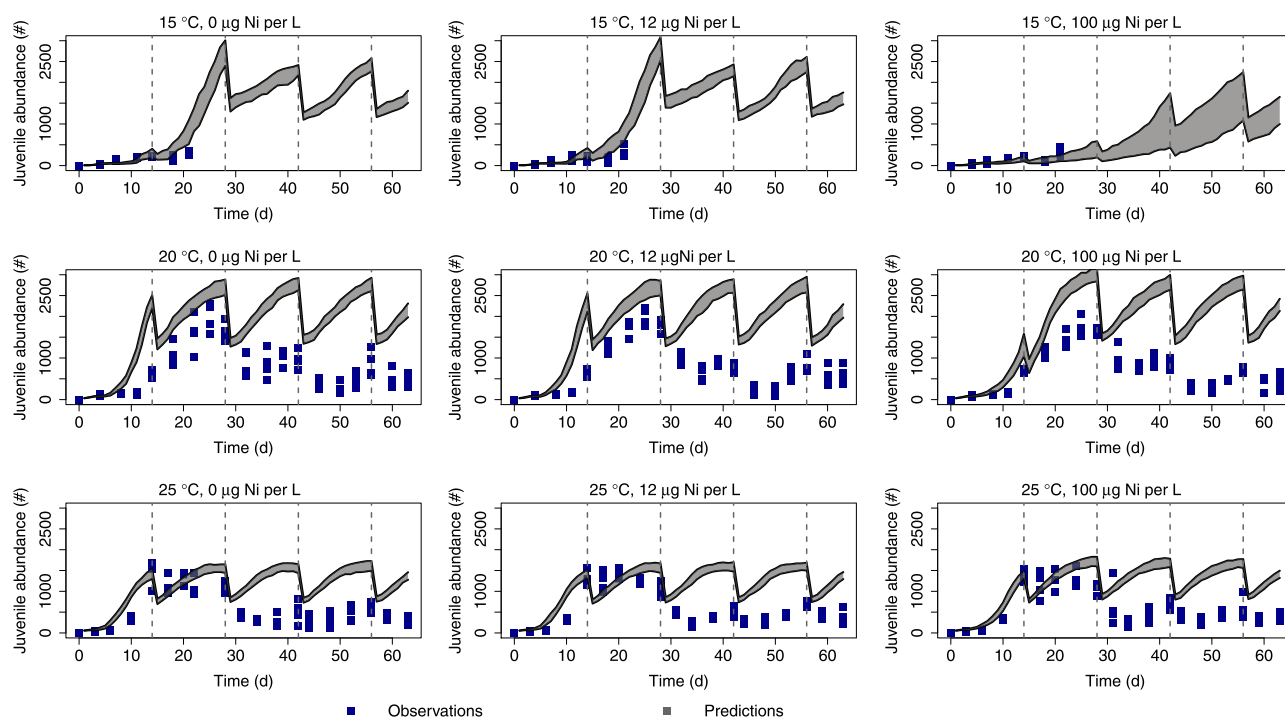


FIGURE 7: Observed and predicted population dynamics for *Daphnia magna* under 0, 12, and 100 μg nickel (Ni) L^{-1} for 9 wk at 15 (data available only until 21 d), 20, and 25 °C. Dots represent data points of the juveniles (and neonates) abundance and lines, the predictions of the model using the physiological mode of action mode with an effect on growth costs (an increase in costs for growth). Vertical lines represent the days (every 14 d) that 50% of each of the 3 size classes (i.e., neonates, juveniles, and adults) were removed at random to avoid population collapses attributable to overcrowding.

difference at 15, 20, and 25 °C, respectively). This difference in Ni effects between the apical and population levels is most likely caused by compensating mechanisms that occur in the population context, where individuals compete with each other, which is not the case in the standard *D. magna* reproduction test. If more individuals in the population die because of Ni exposure, more food will remain available for the surviving, less sensitive individuals. In line with this hypothesis, we performed a detailed analysis of the model to better understand the predicted (absence of) effects at the population level (Supplemental Data, Section 4). We investigated the different causes of mortality with increasing Ni concentration, as well as the changes in DEB energy fluxes over time. We predicted with the DEB-IBM, by tracking in a simulation the cause of death of each individual, that when Ni is present most mortality is caused by Ni, whereas in the control treatment this is predicted to be starvation. Investigating the DEB energy fluxes in detail, the DEB-IBM indeed predicts that a compensation mechanism occurs between food limitation and Ni toxicity. Because more individuals die at intermediate Ni concentrations, more food will be available for the survivors, indicating that they are less limited by food. More energy can then be invested in reproduction, compensating—at the population level—for the Ni-induced mortality. Barata et al. (2002) have experimentally confirmed the compensation of food limitation by toxic effects through a series of population experiments with cladocerans. They observed that toxic stress compensates the stress of the lack of food. Reduction in population

growth rate (by an increase in mortality attributable to toxic stress) caused by a toxicant is compensated because more food is available for survivors, increasing resource efficiency. Similar outcomes have been reported for fruit fly populations (Gui and Grant 2008). In conclusion, because of this compensation mechanism, caused by the joint effect of Ni stress and

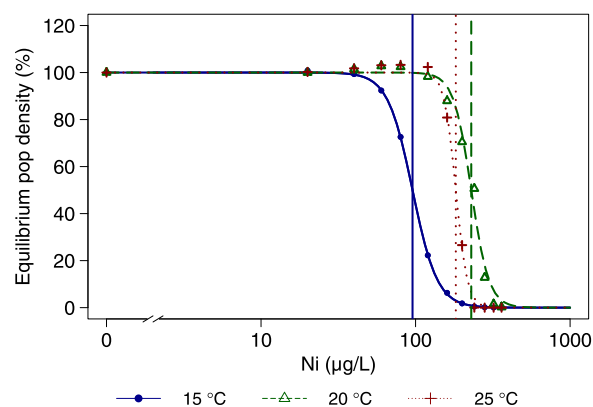


FIGURE 8: The extrapolated concentration–response curves of nickel (Ni) to *Daphnia magna* population abundance (i.e., using the average of the total population abundance from week 8 to week 9) at 15, 20, and 25 °C. Marker points represent the average of the predictions, and lines are the fitted concentration–response curves. Concentration–response curves were fitted through the results of the simulations with the *drc* package (LL.3, 3-parameter log-logistic curve) in R. Vertical lines indicate the extrapolated median effect concentration values at 15, 20, and 25 °C. pop = population.

TABLE 1: Extrapolated effect concentrations of nickel to a *Daphnia magna* population^a

ECx (μg L ⁻¹)	15 °C	20 °C	25 °C
EC10	63.7 (± 0.2)	171.9 (± 0.8)	151.0 (± 0.3)
EC20	74.0 (± 0.2)	191.2 (± 0.6)	161.8 (± 0.2)
EC50	95.6 (± 0.2)	229.3 (± 0.4)	182.1 (± 0.2)

^aUsing the average of the total population abundance from week 8 to week 9; ± standard error) at 15, 20 and 25 °C.

ECx = x% effect concentration.

resource limitation, *D. magna* is less sensitive to Ni at the population level (regardless of the temperature).

The present study together with Viaene et al. (2015) and Gagneten and Vila (2001) show that care should be taken when extrapolating directly from apical to population-level endpoints. Population experiments are time-consuming and expensive, but the use of population models as DEB-IBM can help the scientific community in replacing or optimizing the population experiments. In the present study, the calibrated DEB-IBM predicted the unexpected absence of Ni effect to a *D. magna* population at the tested concentrations. Furthermore, we were able to use the calibrated DEB-IBM to extrapolate the ECx for a *D. magna* population of Ni. This type of information would have been very useful to optimize the experiment design, for example, choosing test concentrations. It has been shown that the DEB-IBM can be calibrated to different test species and different environmental conditions to extrapolate stressor effects to the population level (Martin et al. 2013b; Jager et al. 2014; Hochmuth 2016; Viaene 2016; Ashauer and Jager 2018). For future population studies the prior use of the DEB-IBM to estimate population-level effects of chemical stressors would be helpful to improve the design of the experiment and therefore to optimize the time and the financial resources.

CONCLUSION

The present study showed the absence of consistent Ni effects on a *D. magna* population at 15, 20, and 25 °C at Ni concentrations previously reported to significantly reduce the apical endpoint of reproduction. These results caution that extrapolating from standard toxicity test endpoints to population-level endpoints may be inaccurate. Interestingly, the simplified DEB-IBM calibrated using apical Ni toxicity data at 15, 20, and 25 °C (Pereira et al. 2018) predicted this unexpected absence of Ni effect to a *D. magna* population. Detailed simulations suggest that the lower sensitivity at the population level is most likely attributable to ecological compensation between Ni stress and resource limitation stress (related to intraspecific competition taking place in a population context). Furthermore, the extrapolated EC50 values (predicted with the DEB-IBM) suggest that the effect of temperature on Ni toxicity to *D. magna* population density is considerably smaller than the effect of temperature on Ni toxicity to *D. magna* reproduction. The DEB-IBM can help to optimize the

experimental design of population studies or even to replace population experiments by in silico simulations.

Supplemental Data—The Supplemental Data are available on the Wiley Online Library at DOI: 10.1002/etc.4407.

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Data Accessibility—Data pertaining to the present study are deposited in figshare.

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